

Production-Quality Interactive Course Labs, Built with AI



Tim Fraser, Ph.D.

Assistant Teaching Professor, Systems Engineering

Cornell University · tmf77@cornell.edu
SYSEN 5470 · timothyfraser.com/netsci

11 interactive labs

21 project datasets

2 languages, one lesson

17 packages, zero installs

\$0 servers or hosting

THE PROBLEM

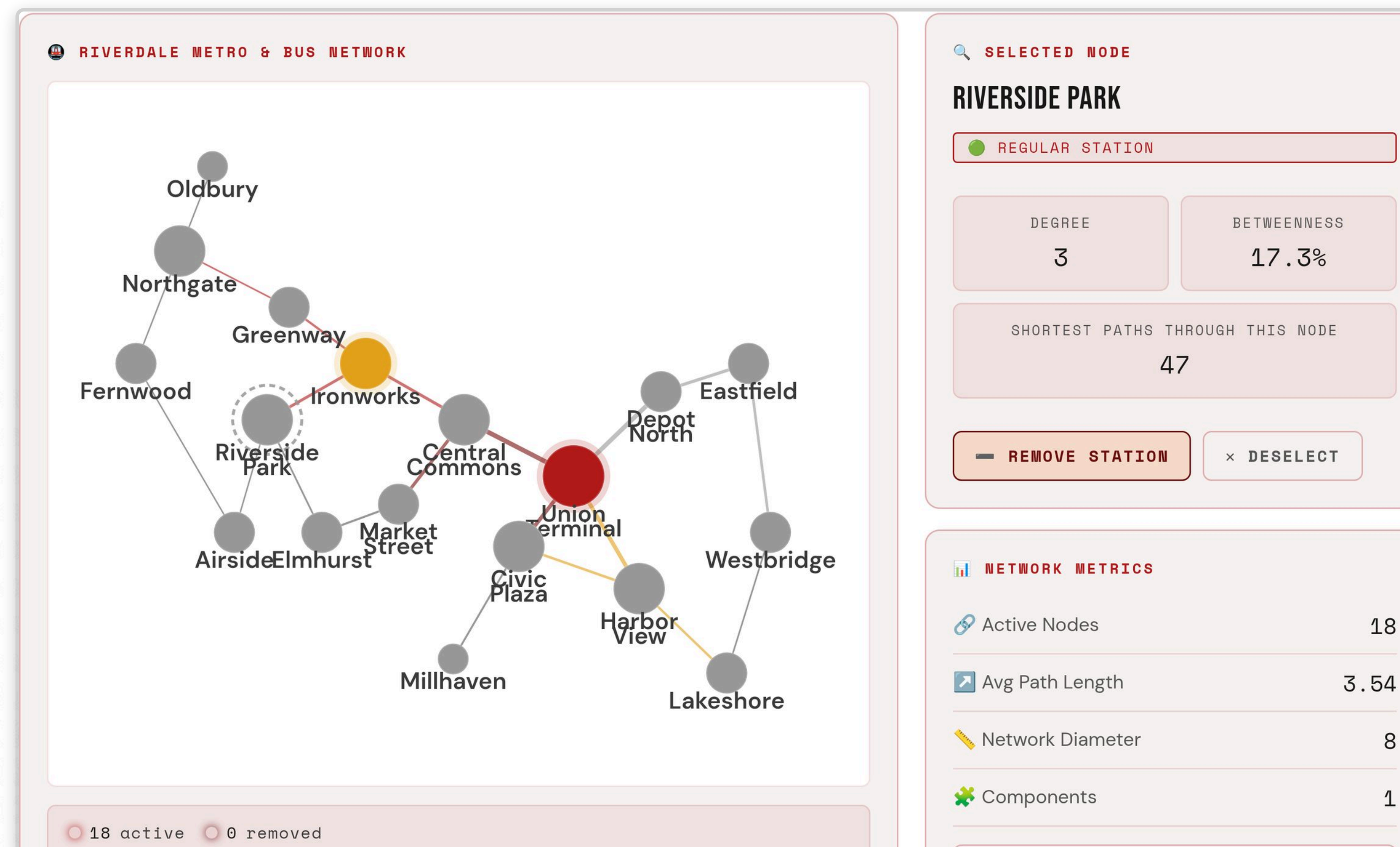
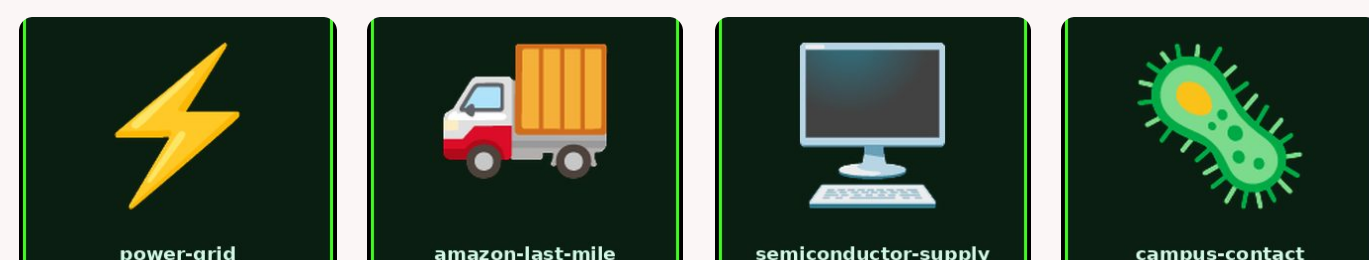
Students thrive with hands-on interactive exercises. But one production-quality lab is weeks of web engineering.

THE AI WORKFLOW

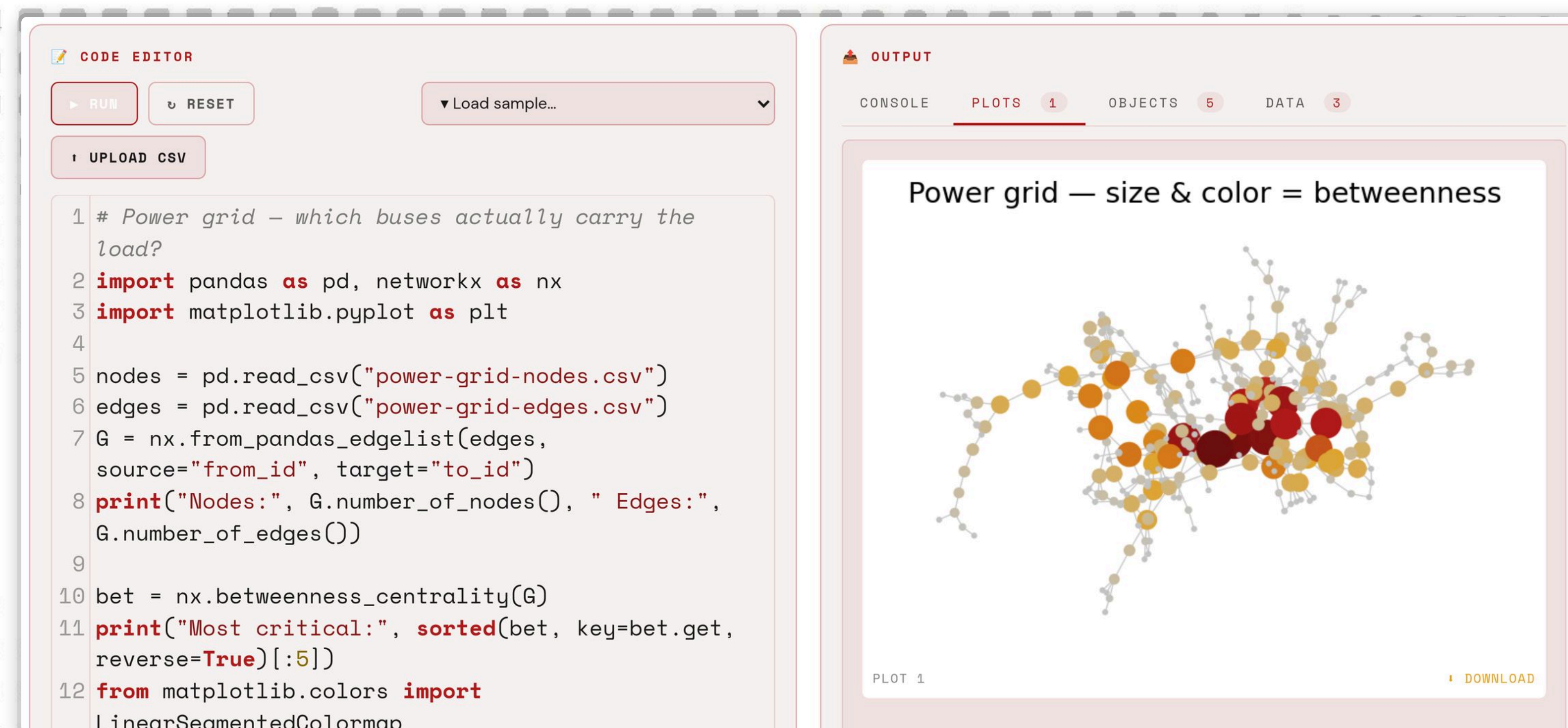
- 1 Encode rules.** Teaching style, glossary, dataset standards.
- 2 Draft with AI.** A lab, a dataset, or lesson code.
- 3 Verify automatically.**
- 4 Instructor reviews.** Design, features, tone, and accuracy — before anything ships.
- 5 Ship free with GitHub Pages.**

21 DATASETS

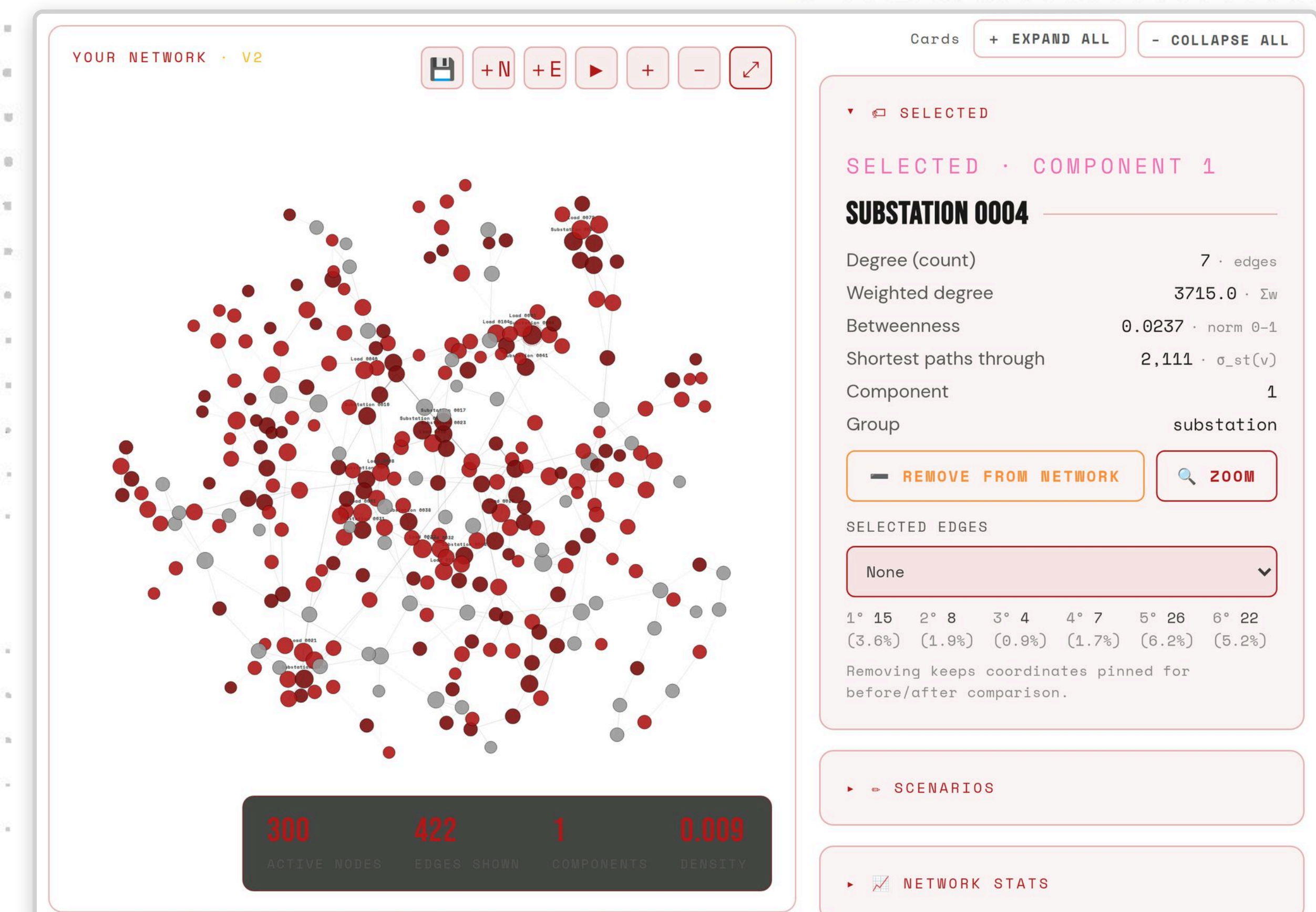
Each hides a planted story, with R + Python loaders.



Case-study lab. Remove a station and every metric updates live.



Coding playground. A full R or Python interpreter in the browser tab, via WebAssembly.



Network Visualizer. Click through failure cascades and counterfactuals on a 300-node grid.

WHAT STUDENTS GET

- **Real code, no setup.** R and Python in the browser.
- **Labs that push back.** Metrics move as you edit.
- **Explore, then code.** Export the script you clicked.
- **A tutor that asks.** Socratic AI, never answers.

TRY IT RIGHT NOW



Course site



Network Visualizer